Financial KPI Analysis for a Startup

Internship Project #7 | Elevate Labs (MSME) | Domain: Data Analytics | Dataset: simulated 12-month financials of an early-stage SaaS startup (Jan-Dec 2025)

1. Introduction

Early-stage startups fail less from lack of ideas than from running out of cash before unit economics prove out. Founders and investors therefore track a small set of financial KPIs every month: revenue (MRR), burn rate, customer acquisition cost (CAC), customer lifetime value (LTV), run rate and cash runway. This project builds an analytics model and dashboard that computes these KPIs from raw monthly financials, tracks their trends, and uses cohort analysis to show whether customers actually stay.

2. Abstract

Twelve months of revenue, expense and customer data were modelled for a subscription startup. MRR grew from **$7.4K to $63.2K** (run rate **$758K**), while net burn peaked at $50.6K in June and fell to **$40.7K** by December. CAC dropped from **$333 to $217** as LTV rose to **$1,096**, lifting the LTV:CAC ratio from **1.7 to 5.1**, past the 3:1 benchmark in June. Cohort analysis shows about 72% of the January cohort still active after five months. With $332K cash, the startup has roughly **8 months of runway**. Outputs: KPI dashboards, an LTV:CAC analysis and a reusable data file.

3. Tools Used

Tool	Purpose
Excel / CSV	Data collection, KPI formulas (burn, CAC, LTV, run rate, runway), reusable model template
Python (Pandas, NumPy)	Data cleaning, KPI computation, cohort retention matrix, trend-based break-even projection
Matplotlib	Dashboard visuals: KPI tiles, trend lines, donut chart, cohort heatmap
Tableau (optional)	The same CSV can be loaded to publish the dashboard as an interactive workbook

Dashboard 1: Performance Overview



Figure 1: Headline KPIs, revenue vs expenses vs burn, customer growth with churn, and Dec expense mix.

4. Steps Involved in Building the Project

- **Collect financials:** built a monthly dataset of new customers, active customers, ARPU, COGS, Sales & Marketing, R&D, G&A and opening cash ($900K). Data is simulated because real startup books are confidential.
- **Clean and validate:** checked for missing months, negative values and consistency (customers = previous x (1 - churn) + new).
- **Compute KPIs:** MRR = customers x ARPU; Net burn = total expenses - revenue; Run rate = MRR x 12; CAC = S&M spend / new customers; LTV = ARPU x gross margin (78%) / churn; Runway = cash / net burn.
- **LTV:CAC ratio:** computed monthly and compared against the 3:1 benchmark.
- **Cohort analysis:** grouped customers by acquisition month (Jan-Jun) and tracked retention for months M0-M5.
- **Dashboard and insights:** built KPI tiles, trend charts and the cohort heatmap, then derived recommendations.

Dashboard 2: Unit Economics, Retention and Cash

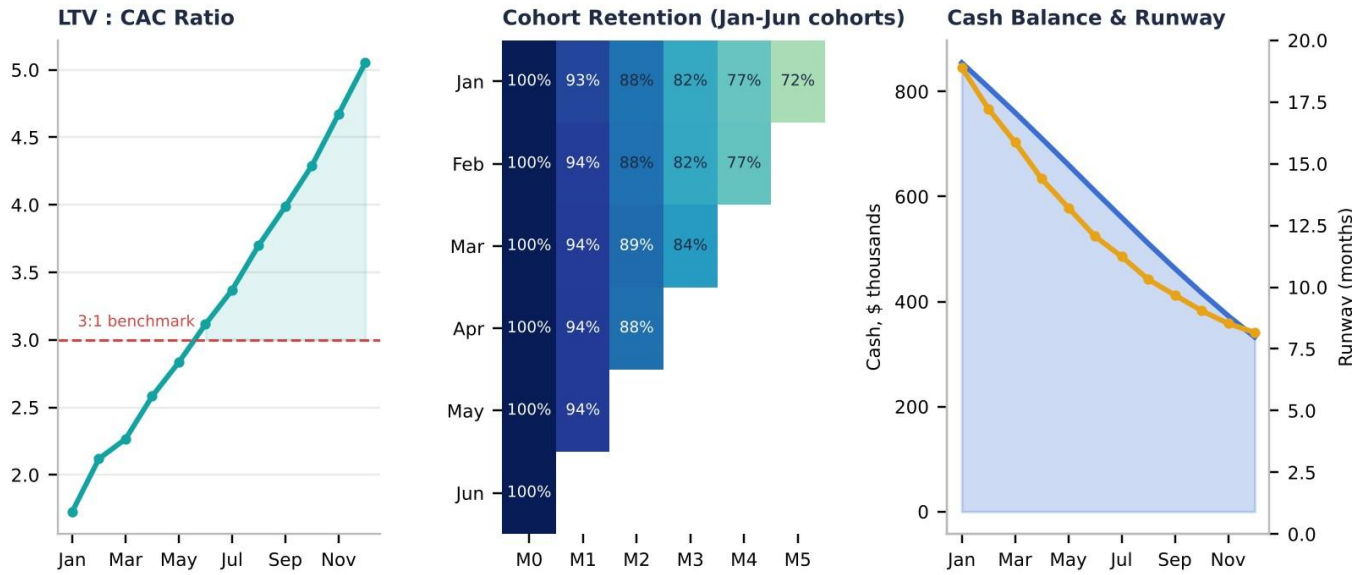


Figure 2: LTV:CAC trend, monthly cohort retention heatmap, and cash balance with runway.

Key Insights

- **Growth is efficient:** MRR grew 8.5x while S&M spend rose only 2.6x, so CAC fell 35% and LTV:CAC crossed 3:1 in June and reached 5.1 by December.
- **Burn is turning down:** burn peaked at $50.6K (Jun) and has fallen for six months because revenue is growing faster (+16% per month) than expenses (+5.6%).
- **Retention needs work:** monthly churn improved from 6.5% to 4.2%, but the Jan cohort kept only ~72% by month 5. Retention is the biggest lever on LTV.
- **Cash is adequate, not comfortable:** 8 months of runway; at current trends revenue covers expenses in about 6 months, leaving roughly $234K cash, a thin but positive margin.

5. Conclusion

The startup shows healthy and improving unit economics: low and falling CAC, rising LTV, declining burn and a clear path to break-even within about 6 months on current trends. The main risks are customer churn and a limited cash buffer. Recommended actions: (1) invest in onboarding and retention to lift LTV further, (2) keep S&M spend growth below revenue growth, and (3) begin the next fundraise or secure a bridge while runway is still above six months. The Python/Excel pipeline is reusable: replacing the CSV with real monthly financials refreshes every KPI and chart. *Note: figures come from a simulated dataset; they illustrate the method, not a real company.*